

Porosity-specific particularities of the OSGS a priori analysis

A term-by-term comparison with Codina’s orthogonal-subscale analysis of the Oseen equations, and where the two proofs diverge

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Abstract

This note compares, assumption by assumption and proof step by proof step, the orthogonal-subgrid-scale (OSGS) a priori analysis of the linearized incompressible inertial porous-media flow equations—the analysis carried out in the companion manuscript [1] and its self-contained backing note [2]—with its direct ancestor, Codina’s analysis of the Oseen equations by orthogonal subscales [3]. Three questions are addressed. First, *what is forced by the new equation*: the porosity weight $\alpha(\mathbf{x})$, the Darcy reaction $\sigma \mathbf{u}$, the artificial-compressibility term εp , and the deviatoric-symmetric viscous operator Π^{DS} each dictate concrete changes to the hypotheses, the stabilization parameters, the working norm and the estimates, catalogued in Section 4. Second, *where the two proofs differ as proofs*, summarized as a step-by-step ledger in Section 6. Third—answering an explicit question, and with the honest qualification that the answer is mostly “no error, but the exposition can be tightened”—*where the Oseen analysis of [3] is less than fully explicit*, and how the porous analysis sharpens it (Section 5). Of the points examined, exactly one (the smoothing lemma silently requiring a smooth-grading mesh hypothesis) is a genuine, if benign, implicit assumption; the others are matters of *sharpness* (the ℓ^1 error functional), *disclosure* (stabilization constants), or *cleaner devices* (variable- τ projection handling). None is a soundness defect: every theorem of [3] is correct as stated, and R. Codina is a co-author of [1]. Throughout, “Codina 2007” denotes the 2007-online / 2008-print paper [3] (Appl. Numer. Math. **58** (2008) 264–283); the companion note [2] cites it under the key **Codina2008**.

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1 Scope and the correct triangulation

Codina 2007 [3] analyzes the stationary *Oseen* problem

$$-\nu\Delta\mathbf{u} + \mathbf{a} \cdot \nabla\mathbf{u} + \nabla p = \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0 \quad \text{in } \Omega, \quad \mathbf{u} = \mathbf{0} \quad \text{on } \partial\Omega, \quad (1)$$

stabilized by orthogonal subscales. It contains *three* analyzed formulations: **Method I** (a single projected residual per equation; Theorems 1–2), **Method II** (convective and pressure-gradient fluctuations controlled separately, in a finer norm; Theorems 3–4), and a **pressure-only, viscous-dominated variant** whose norm depends explicitly on the mesh and global Reynolds numbers (Theorems 5–6). The companion paper [1, 2] treats the linearized porous problem

$$\alpha \mathbf{a} \cdot \nabla\mathbf{u} - 2\nabla \cdot (\alpha\nu \Pi^{\text{DS}}\nabla\mathbf{u}) + \alpha\nabla p + \sigma\mathbf{u} = \mathbf{f} \quad \text{in } \Omega, \quad (2a)$$

$$\varepsilon p + \nabla \cdot (\alpha\mathbf{u}) = 0 \quad \text{in } \Omega, \quad \mathbf{u} = \mathbf{0} \quad \text{on } \partial\Omega, \quad (2b)$$

and adapts **only Method I** (Section 6 records this as the single largest scope difference). Method II survives in [2] as a one-paragraph prospective remark whose proof is not written out; the viscous-dominated pressure-only variant has no porous analogue at all.

Two ancestor papers, two routes. A recurring source of confusion must be dispatched at the outset. The porous work has *two* a priori analyses following *two different* Codina papers along *two different* routes:

Porous variant	Template	Route
ASGS (companion App. C [1])	Codina 2001 [4]	coercivity / Céa
OSGS (companion App. D [1], = [2])	Codina 2007 [3]	non-coercive inf-sup

This note compares the OSGS row only: Codina 2007 versus the porous OSGS analysis. The consequence is important for a fair comparison of assumptions. Objects such as the interior-face jump condition on τ_1 , the enlarged inverse constant $\bar{C}_{\text{inv}}$ for $\nabla \cdot (\alpha\nu \Pi^{\text{DS}}\nabla\mathbf{u}_h)$, and the damped reaction coefficient $\tilde{\sigma}_\alpha$ are *Codina 2001* $\leftrightarrow$ *porous-ASGS* objects. Codina 2007 is pure Oseen ($\sigma = 0$) and has none of them; the porous OSGS analysis *dispenses* with the first two relative to the porous ASGS analysis and never had the third. It would therefore be wrong to say “the OSGS variant drops hypothesis H:jump of Codina 2007”—that hypothesis is not in Codina 2007.

A symbol collision to keep in view. Codina 2007 uses the letters α and σ for *design constants*: $\alpha > 1$ is the stabilization-amplification factor in $c_1 = \alpha^2 C_{\text{inv}}^2$, $c_2 = \alpha C_{\text{inv}}$, and $0 < \sigma \leq 1$ appears in $c_3 = \sigma$, $c_4 = \sigma/(\alpha C_{\text{inv}})$ (his Eqs. (27)–(28)). In the porous problem the same letters are *physical fields*: α is the porosity and σ the Darcy reaction. The porous note therefore renames Codina’s amplification factor $\alpha \rightarrow \gamma$ (hypothesis (A6): $c_1 \geq \gamma^2 C_{\text{inv}}^2$, $c_2 \geq \gamma C_{\text{inv}}$), and Codina’s design constant σ simply disappears because, as Section 4.5 shows, the porous τ_2 is not free. *One must not identify the porous α, σ with Codina’s α, σ .*

What is reused verbatim. The porous OSGS proof reuses from Codina 2007, essentially unchanged in *structure*: the non-coercive inf-sup argument built on a specially designed test function ($\tau \otimes \Pi_0$ of the projected residual); the continuous element-size surrogate of his Lemma 1; the smoothing operation of his Lemma 2; the best-approximation property of the weighted projection (his Remark 4, Eq. (46)); and the “interpolate-consistency-triangle” skeleton of his Theorem 2. What changes is dictated, almost entirely, by the four new ingredients of Equation (2).

2 The two model problems, side by side

Table 1 isolates the four structural differences between Equation (1) and Equation (2). Everything in Sections 4 to 6 is a consequence of one of these four rows.

Ingredient	Codina 2007 (Oseen)	Porous linearized NS
Viscous operator	$-\nu\Delta\mathbf{u}$; energy $\nu\ \nabla\mathbf{u}\ ^2$	$-2\nabla \cdot (\alpha\nu\Pi^{\text{DS}}\nabla\mathbf{u})$; energy $2\nu\ \alpha^{1/2}\Pi^{\text{DS}}\nabla\mathbf{u}\ ^2$
Coefficient weight	unweighted operators; $\nabla \cdot \mathbf{a} = 0$	everything weighted by porosity $\alpha(\mathbf{x})$; weighted solenoidality $\nabla \cdot (\alpha\mathbf{a}) = 0$
Zero-order term	none	Darcy reaction $\sigma\mathbf{u}$, $\sigma \geq 0$ constant
Mass equation	$\nabla \cdot \mathbf{u} = 0$ (exactly incompressible)	$\varepsilon p + \nabla \cdot (\alpha\mathbf{u}) = 0$ (artificial compressibility $\varepsilon \geq 0$)

Table 1: The four ingredients that separate the porous problem from Oseen. The Galerkin form gains, respectively, a deviatoric–symmetric viscous pairing, the porosity weight, the reaction pairing $\sigma(\mathbf{v}, \mathbf{u})$, and the compressibility pairing $\varepsilon(q, p)$.

The Galerkin bilinear form and its coercivity (“diagonal”) identity make the four additions concrete. In Codina 2007 (his Eqs. (4), (39)),

$$B(U, V) = \nu(\nabla\mathbf{u}, \nabla\mathbf{v}) + (\mathbf{a} \cdot \nabla\mathbf{u}, \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) + (q, \nabla \cdot \mathbf{u}), \quad B(U, U) = \nu\|\nabla\mathbf{u}\|^2;$$

in the porous problem ([2], Eqs. GalerkinForm, CoercivityIdentity),

$$B(\mathbf{a}; U, V) = (\mathbf{v}, \alpha\mathbf{a} \cdot \nabla\mathbf{u}) + 2(\nabla\mathbf{v}, \alpha\nu\Pi^{\text{DS}}\nabla\mathbf{u}) + (\mathbf{v}, \alpha\nabla p) + \sigma(\mathbf{v}, \mathbf{u}) + \varepsilon(q, p) + (q, \nabla \cdot (\alpha\mathbf{u})),$$

$$B(\mathbf{a}; U, U) = 2\nu\|\alpha^{1/2}\Pi^{\text{DS}}\nabla\mathbf{u}\|^2 + \|\sigma^{1/2}\mathbf{u}\|^2 + \varepsilon\|p\|^2.$$

The two *new* diagonal quantities $\|\sigma^{1/2}\mathbf{u}\|^2$ and $\varepsilon\|p\|^2$, and the replacement of $\nu\|\nabla\mathbf{u}\|^2$ by the deviatoric–symmetric $2\nu\|\alpha^{1/2}\Pi^{\text{DS}}\nabla\mathbf{u}\|^2$, propagate into every downstream estimate.

3 Assumptions: H1–H6 versus (A1)–(A8)

Table 2 aligns Codina’s six hypotheses with the porous eight and classifies each porous item as F (forced by the new physics), T (a tightening or explicit disclosure of something implicit in Codina 2007), or an inherited transcription.

Two observations deserve emphasis.

Observation 1 (The pressure-regularity/order refinement). Codina’s H6 asks only for $p \in H^k(\Omega)$ with a *single* interpolation degree k , and his error function carries the pressure at order h^k —one order below the velocity’s h^{k+1} . The porous (A8) asks for $p \in H^{k_p+1}$ and carries the pressure at its own *full* order h^{k_p+1} , with a possibly distinct degree $k_p \leq k_u + 1$. This is a genuine hypothesis and error-order difference, orthogonal to the four physical ingredients, and it is worth stating explicitly because it changes what “optimal” means for the pressure.

Observation 2 (The reaction is not wholly unprecedented in Codina’s programme). Codina 2007 itself notes (immediately after its Eq. (16)) that reference [4]—the very paper the porous *ASGS* analysis follows—already analyzed “zero-order terms coming from Coriolis forces and permeability

Porous	Content	Relation to Codina 2007
(A1)	Nondegenerate mesh; inverse estimate; quasi-uniform patches; continuous element-size surrogate.	= H3 + his Lemma 1. Inherited.
(A2)	$\nu > 0$, $\sigma \geq 0$, $\varepsilon \geq 0$ constant; Darcy tensor <i>isotropic</i> , $\sigma = \sigma \mathbf{1}$.	F. No Codina analogue; restricts the paper's general Forchheimer tensor $\sigma(\alpha, \mathbf{u})$ so that the annihilation lemma holds.
(A3)	Porosity $\alpha \in W^{1,\infty}$, $0 < \alpha_0 \leq \alpha \leq \alpha_\infty$ (≤ 1); <i>resolution</i> $h_K \ \nabla \alpha\ _{\infty, K} \leq C_{\nabla \alpha} \alpha_K$; and $\alpha \in W^{m,\infty}$, $m = \max(k_u, k_p)$, when $m \geq 2$.	F. Entirely new. Codina has $\alpha \equiv 1$.
(A4)	$\mathbf{a} \in C^0$, elementwise $W^{1,\infty}$; <i>weighted</i> solenoidality $\nabla \cdot (\alpha \mathbf{a}) = 0$; and $\ D^j \mathbf{a}\ _{\infty, K} \leq C_D \mathbf{a} _{\infty, K}$ for consistency.	F (weight) + = H1 (regularity) + = H2 (derivative bound). Codina: $\nabla \cdot \mathbf{a} = 0$.
(A5)	Continuous V_h, Q_h of degrees $k_u, k_p \geq 1$, $k_p \leq k_u + 1$; constants in Q_h ; Dirichlet data.	Refines Codina's "degree k for both": the porous analysis carries <i>separate</i> k_u, k_p . T.
(A6)	$c_1 \geq \gamma^2 C_{\text{inv}}^2$, $c_2 \geq \gamma C_{\text{inv}}$, $\gamma \geq \gamma_0$; and $\varepsilon \tau_{2,K} \leq C_2 \leq 1$.	= H5 with amplification $\alpha \rightarrow \gamma$; <i>explicit</i> admissible γ_0 (Codina leaves " α large enough" qualitative). ε -bound is new (F by ε). T/F.
(A7)	Projection stability: $\ z_h\ _{\tau_1} \leq \beta_0^{-1} (\ \Pi_{1,0} z_h\ _{\tau_1} + \ \Pi_1^\perp z_h\ _{\tau_1})$ for $z_h = \alpha X(V_h)$.	= H4 (his condition (35)), transported to the <i>weighted</i> residual, and imposed only on the <i>momentum</i> slot (see Section 6).
(A8)	$\mathbf{u} \in H^{k_u+1} \cap H_0^1$, $p \in H^{k_p+1} \cap Q_0$.	T. Stronger than H6, which assumes only $p \in H^k$ (one order below velocity).

Table 2: The porous hypotheses against Codina's. F: forced by the new physics. T: a tightening/disclosure of something Codina left implicit or coarser.

effects”, i.e. an algebraic-subscale analysis *with* a permeability zero-order term. Thus the Darcy reaction is unprecedented specifically in the *orthogonal*-subscale setting of Codina 2007, not in Codina’s stabilized-FEM programme as a whole. The novelty below is precise: it is the interaction of the reaction with the *orthogonal projection* (the annihilation of the reactive residual) that is new.

4 How the new equation forces the differences

We take the four ingredients in turn, tracing each to the assumptions, parameters, norm, and estimates it dictates.

4.1 Porosity weight $\alpha(\mathbf{x})$

The porosity does not merely rescale; because it is a *variable field* it changes the algebra of every step.

1. *The stabilized residual carries the weight.* The natural stabilized quantity is $\alpha X(V) = \alpha(\mathbf{a} \cdot \nabla \mathbf{v} + \nabla q)$, not the bare $X(V)$ of Codina. Both stabilization terms S_1, S_2 ([2], Eq. Sdefs) project the *weighted* momentum residual $\alpha X(U)$ and the *weighted* divergence $\nabla \cdot (\alpha \mathbf{u})$.
2. *Weighted solenoidality.* The convective skew-symmetry that makes the Galerkin form diagonal now requires $\nabla \cdot (\alpha \mathbf{a}) = 0$ (assumption (A4)): $(\mathbf{u}, \alpha \mathbf{a} \cdot \nabla \mathbf{u}) = \frac{1}{2}(\alpha \mathbf{a}, \nabla |\mathbf{u}|^2) = -\frac{1}{2}(\nabla \cdot (\alpha \mathbf{a}), |\mathbf{u}|^2) = 0$. Codina’s $\nabla \cdot \mathbf{a} = 0$ (H1) is the $\alpha \equiv 1$ special case.
3. *Porosity-resolution condition (A3).* A hypothesis with *no* Codina analogue: $h_K \|\nabla \alpha\|_{\infty, K} \leq C_{\nabla \alpha} \alpha_K$. It is consumed in three places: (i) a dedicated *multiplicative* patch-equivalence lemma for α ([2], lem:patch), whose constant $\delta_\alpha^2 = (1 + C_{\nabla \alpha})^2$ depends only on $C_{\nabla \alpha}$ and *not* on the porosity contrast α_∞/α_0 ; (ii) the $\nabla \alpha$ term that appears whenever $\nabla \cdot (\alpha \mathbf{w}) = \alpha \nabla \cdot \mathbf{w} + \nabla \alpha \cdot \mathbf{w}$ is expanded (stability term T7, interpolation term I7); (iii) the Leibniz step in the weighted best-approximation lemma. Codina’s Lemma 1 equilibrates only τ across a patch; it never equilibrates a *coefficient* like α .
4. *The $\nabla \alpha$ term inflates the stability threshold.* Because $\nabla \cdot (\alpha \mathbf{w}_h)$ carries the extra $\nabla \alpha \cdot \mathbf{w}_h$ piece, term T7 gains the factor $M_2 = 1 + C_{\nabla \alpha}/C_{\text{inv}}$, which propagates into the admissible design constant $\gamma_0 = 10(1 + M_2)$. In Codina, where $\alpha \equiv 1$, the corresponding term has only $\nabla \cdot \mathbf{w}_h$ and the threshold is merely “ $\alpha > 3/2$ ”.
5. *Weighted inverse estimates.* Codina’s three closing inequalities on his p. 273 gain the weight α_K and become (K1)–(K3) of [2] (e.g. $C_{\text{inv}}(\alpha_K \nu)^{1/2}/h_K \leq \gamma^{-1} \tau_1^{-1/2}$), with the amplification $\alpha \rightarrow \gamma$ already noted.
6. *Extra porosity smoothness for $k \geq 2$.* The best-approximation lemma must project the *product* αz ; a Leibniz bound $\|\alpha z\|_{H^r(S_K)} \leq C C_{\alpha, r} \alpha_\infty \|z\|_{H^r(S_K)}$ forces the constant $C_{\alpha, r}$ and hence the $W^{m, \infty}$ tail of (A3). Codina projects z itself.

4.2 Darcy reaction $\sigma \mathbf{u}$: the structural heart

This is where the analysis genuinely departs from Codina 2007, through a single mechanism.

Annihilation. For constant σ , $\sigma \mathbf{u}_h$ is a finite element function, so its orthogonal fluctuation vanishes identically: $\Pi_1^\perp(\sigma \mathbf{u}_h) = \mathbf{0}$ ([2], lem:annihilation). The orthogonal projection therefore *deletes* the reactive residual. There is no reactive subscale to renormalize the reaction coefficient.

σ versus $\tilde{\sigma}_\alpha$ (the trade). Contrast the ASGS variant, whose residual-based stabilization retains the reactive fluctuation and produces the diagonal $\sigma(1 - \sigma\tau_1)\|\mathbf{u}_h\|^2 = \tilde{\sigma}_\alpha\|\mathbf{u}_h\|^2$ with the *damped*

$$\tilde{\sigma}_\alpha = \frac{\tau_{1,\text{NS}}^{-1}\sigma}{\tau_{1,\text{NS}}^{-1} + \sigma/\alpha_K} \leq \sigma.$$

Because OSGS annihilates the reactive residual, no such damping occurs: the *full* $\|\sigma^{1/2}\mathbf{u}_h\|^2$ survives in the diagonal and in the working norm. This is a strictly *stronger* stability estimate—the porous OSGS triple norm dominates the ASGS triple norm term by term ([2], `lem:normcomparison`). The price is paid on the other side of the ledger: the reactive and pressure *interpolation* errors, no longer absorbable through a reactive subscale, must be bounded head-on, and the best conversions available cost a Damköhler factor.

The Damköhler factor and its two new identities. The reaction enters τ_1 through its denominator, $\tau_1 = (\alpha_K\tau_{1,\text{NS}}^{-1} + \sigma)^{-1}$ (Codina’s τ_1 has no reaction), and this forces two parameter inequalities with no Codina analogue, in terms of the elemental *mesh Damköhler number* $\text{Da}_{h,K} = \sigma h_K^2/(\alpha_K\nu)$:

$$(P5) \quad \alpha_K^2\tau_1\tau_2 \geq \frac{h_K^2}{c_1 + \text{Da}_{h,K}}, \quad \text{hence} \quad \tau_2^{-1/2} \leq (c_1 + \text{Da}_{h,K})^{1/2} \frac{\alpha_K}{h_K} \tau_1^{1/2}, \quad (3)$$

$$(P6) \quad \sigma^{1/2} \leq \text{Da}_{h,K}^{1/2} \frac{\alpha_K}{h_K} \tau_2^{1/2}. \quad (4)$$

(P5) is the sole origin of the $(1 + \text{Da}_{h,K})^{1/2}$ inflation. When $\sigma = 0$ it collapses to Codina’s exact relation “ $\tau_{1,K}$ behaves as $h_K^2\tau_{2,K}^{-1}$ ” (the last line of his Lemma 3 proof), i.e. $\alpha_K^2\tau_1\tau_2 = h_K^2/c_1$. The reaction thus *adds* $\text{Da}_{h,K}$ to a quantity that was a clean equality in the Oseen case.

New proof terms. The reaction produces the new stability pairing $\text{T3} = \sigma(\mathbf{u}_h, \mathbf{w}_h)$, absorbed by the full diagonal $\|\sigma^{1/2}\mathbf{u}_h\|^2$ using only $\sigma\tau_1 \leq 1$; the new interpolation group I3; the reaction-inflated error functional (Section 4.5); and an entirely new corollary—the σ -robust optimal L^2 velocity estimate ([2], `cor:Ltwo`), obtained by dividing the σ -controlled triple norm by $\sigma^{1/2}$, with the pressure contribution *decaying* as $(\sigma\nu)^{-1/2}$. Codina has no reaction and therefore no L^2 velocity control except in his Reynolds-degrading viscous-dominated variant.

4.3 Artificial compressibility εp : a new consistency term

The compressibility is easy to overlook but produces a qualitatively new phenomenon.

1. *New diagonal and new pairing.* The form gains $\varepsilon(q, p)$; the diagonal gains $\varepsilon\|p_h\|^2$; the stability expansion gains $\text{T4} = \varepsilon(p_h, r_h)$, controlled by the new design bound $\varepsilon\tau_2 \leq C_2 \leq 1$ (assumption (A6))—an upper bound on ε absent from Codina.
2. *The mass consistency term stops vanishing.* This is the salient point. In Codina, $\nabla \cdot \mathbf{u} = 0$ *exactly*, so the mass part of his consistency error (his Eq. (44), second term) is $(\Pi_2^\perp(\nabla \cdot \mathbf{u}), \cdot) = 0$: it vanishes identically. In the porous problem the exact mass equation reads $\nabla \cdot (\alpha\mathbf{u}) = -\varepsilon p$, so

$$S_2(U, V_h) = -\varepsilon(\Pi_2^\perp p, \nabla \cdot (\alpha\mathbf{v}_h))_{\tau_2} \neq 0 \quad (\varepsilon > 0),$$

a genuinely new consistency contribution that must be bounded head-on, via $\varepsilon \leq C_2\tau_2^{-1}$ and (P5), and contributes a $(c_1 + \text{Da}_{h,K})$ -carrying term to the error functional. It vanishes precisely when $\varepsilon = 0$, isolating ε as its sole driver. There is also a matching interpolation group I4 and the annihilation $\Pi_2^\perp(\varepsilon p_h) = 0$ for constant ε .

4.4 Deviatoric–symmetric viscous operator Π^{DS}

The viscous term is $-2\nabla \cdot (\alpha\nu\Pi^{\text{DS}}\nabla\mathbf{u})$, with Π^{DS} the pointwise orthogonal projection onto the deviatoric–symmetric part of a tensor. Two consequences.

1. *Energy in a projected seminorm.* The diagonal and the norm carry $2\nu\|\alpha^{1/2}\Pi^{\text{DS}}\nabla\mathbf{u}\|^2$ instead of Codina’s $\nu\|\nabla\mathbf{u}\|^2$. The perturbation bound for the viscous cross term (T1, I1) uses $|\Pi^{\text{DS}}\mathbf{M}| \leq |\mathbf{M}|$ pointwise.
2. *Definiteness of the norm is no longer trivial.* Because control is on $\Pi^{\text{DS}}\nabla\mathbf{v}$, not $\nabla\mathbf{v}$, the triple norm is definite only through a Korn-type identity

$$\|\Pi^{\text{DS}}\nabla\mathbf{v}\|^2 = \frac{1}{2}\|\nabla\mathbf{v}\|^2 + \left(\frac{1}{2} - \frac{1}{d}\right)\|\nabla \cdot \mathbf{v}\|^2,$$

valid on H_0^1 (full Dirichlet data, via integration by parts), whence $\|\nabla\mathbf{v}\| \leq \sqrt{2}\|\Pi^{\text{DS}}\nabla\mathbf{v}\|$; absent full Dirichlet data or pressure normalization the triple norm is only a *seminorm* ([2], `lem:definiteness`). Codina’s $\nu^{1/2}\|\nabla\mathbf{v}\|$ is a norm outright by Poincaré. The companion paper flags that this deviatoric structure is precisely what would make a Korn-type inequality nontrivial for a full existence theory.

3. *The neglected viscous residual carries $\nabla\alpha$ even at $k_u = 1$.* In Codina, the dropped viscous residual $\nu\Delta\mathbf{u}_h$ vanishes on element interiors for linear elements. Here the dropped residual is $2\nu(\Pi^{\text{DS}}\nabla\mathbf{u}_h)\nabla\alpha$, nonzero already at $k_u = 1$ whenever α varies (though small under (A3)). This slightly widens the analyzed-versus-implemented distance of the Method I simplification.

4.5 A parameter re-parametrization worth naming

Codina 2007 uses *four independent* design constants: $\tau_1 = [c_1\nu/h^2 + c_2|\mathbf{a}|/h]^{-1}$ and $\tau_2 = c_3\nu + c_4|\mathbf{a}|h$ (his Eqs. (11)–(12)), with c_1, c_2 in (27) and c_3, c_4 in (28). The porous τ_2 is *not free*: it is slaved to $\tau_{1,\text{NS}}$,

$$\tau_{2,K} = \frac{h_K^2}{c_1\alpha_K\tau_{1,\text{NS}}} = \frac{\nu}{\alpha_K} \left(1 + \frac{c_2}{c_1}\text{Re}_{h,K}\right),$$

so the exact identity $h_K^2\tau_{1,\text{NS}}^{-1} = c_1\alpha_K\tau_2$ (P3) holds, and only c_1, c_2 survive as free constants. This coupling—not present in Codina—is what *generates* (P3) and (P5) and hence the Damköhler factor. The reduction of the porous theorem to Codina’s Theorem 2 therefore requires the dictionary $c_3 = 1$, $c_4 = c_2/c_1$ (in the limit $\alpha \equiv 1$, $\sigma = \varepsilon = 0$); it is not a free relabeling of four constants but a consequence of the two-constant structure.

The error functionals reflect the same story. Codina’s is the ℓ^1 element sum (his Eq. (36))

$$\varepsilon_{\text{C}}(h) = \sum_K \left(\tau_{1,K}^{-1/2} h_K^{k+1} \|\mathbf{u}\|_{k+1,K} + \tau_{2,K}^{-1/2} h_K^k \|p\|_{k,K} \right);$$

the porous theorem is stated in the *sharp* porosity-weighted broken ℓ^2 functional

$$\Psi_{\text{O}}(h) = \left(\sum_K (c_1 + \text{Da}_{h,K}) \frac{\alpha_K^2}{h_K^2} (\tau_{2,K} \text{E}_{\text{int},K}(\mathbf{u})^2 + \tau_{1,K} \text{E}_{\text{int},K}(p)^2) \right)^{1/2},$$

whose coarser ℓ^1 majorant $\mathcal{E}(h)$ (the ASGS error function times $(1 + \text{Da}_{h,K})^{1/2}$) is retained only as a comparator. The relation $\Psi_{\text{O}} \leq c_1^{1/2}\mathcal{E}$ and the deliberate choice to state the theorem in Ψ_{O} is the subject of Section 5 (i).

5 Where the Oseen analysis is less than fully explicit

The question “are there places where the original paper was not completely rigorous?” has, after a careful re-reading, a nuanced answer, and it is worth stating the verdict before the evidence: **Codina 2007 contains no soundness error.** Every inequality is valid and every theorem holds as stated. What the porous analysis does—appropriately, since Codina co-authors it—is (i) state one estimate in a sharper norm, (ii) surface one implicit mesh hypothesis, (iii) disclose the gap between provable and practical constants, and (iv) replace one “trivially” with an exact, variable-coefficient-robust device. Only item (ii) is a genuine (and entirely benign) implicit assumption. We take them in order, each with its fair severity.

(i) The ℓ^1 error functional is coarser than the sharp ℓ^2 object (sharpness, not soundness)

Codina’s convergence Theorems 2 and 4 read $\|U - U_h\| \leq C\varepsilon_C(h)$, with $\varepsilon_C(h)$ the ℓ^1 element sum above. But the τ -norm $\|\cdot\|_\tau = (\sum_K \tau_K \|\cdot\|_K^2)^{1/2}$ that appears throughout the proof is an ℓ^2 object, and the passage from it to $\varepsilon_C(h)$ —his Eq. (46), $\|v - \Pi_\tau v\|_\tau \leq C \sum_K \tau_K^{1/2} h_K^r \|v\|_{H^r(K)}$ —uses the valid but lossy step $(\sum b_K^2)^{1/2} \leq \sum b_K$.

The gap is quantifiable. On a quasi-uniform mesh in d dimensions with $N \sim h^{-d}$ elements, a smooth solution has $\|\mathbf{u}\|_{H^{k+1}(K)} = O(h^{d/2})$ (since $|K| = O(h^d)$). With the diffusion-dominated scaling $\tau_1^{-1/2} \sim \nu^{1/2}/h$, the per-element term is $a_K = O(\nu^{1/2} h^{k+d/2})$; the sharp ℓ^2 functional is $\Psi = (\sum a_K^2)^{1/2} = N^{1/2} a = O(\nu^{1/2} h^k)$ —the *optimal* energy-norm rate—while the literal ℓ^1 sum is $\varepsilon_C = \sum a_K = Na = O(\nu^{1/2} h^{k-d/2})$. The ratio is exactly $N^{1/2} = h^{-d/2}$: for $d = 2, k = 1$, $\Psi = O(h)$ but $\varepsilon_C = O(1)$; for $d = 3, k = 1$, $\varepsilon_C = O(h^{-1/2})$, which does not even go to zero.

Note. This is a matter of *sharpness*, not correctness. The theorem $\|U - U_h\| \leq C\varepsilon_C(h)$ is a true inequality; the optimal rate is genuinely delivered by the ℓ^2 object that already lives inside Codina’s own proof (the τ -norm best-approximation). The $O(h^{k-d/2})$ reading is what one gets by substituting local smooth-solution scalings term by term into the ℓ^1 sum—something no informed reader of the stabilized-FEM literature does, where $\sum_K h^{k+1} \|\mathbf{u}\|_{k+1,K}$ is understood at optimal order. The porous analysis simply makes this explicit: it states its theorem in Ψ_O and *names* the loss (“the ℓ^1 passage to $\mathcal{E}$ can lose a mesh-cardinality factor $\sim h^{-d/2}$ on a quasi-uniform family, so the optimal-order corollaries must be read off the ℓ^2 functional”). Severity: negligible as correctness, moderate as presentation. Note that the porous *ASGS* appendix itself still uses an ℓ^1 error functional in the Codina style, so this refinement is specifically a feature of the *OSGS* note.

(ii) The smoothing lemma silently needs a smooth-grading mesh hypothesis (a genuine, benign gap)

This is the one point at which a hypothesis is genuinely missing. Codina’s Lemma 2 bounds $\|\tau \circ v_h - \tau \otimes v_h\|_K \leq \tau_K \psi(h) \|v_h\|_K$ with $\psi(h) \rightarrow 0$, and its proof reduces this to $\max_{a \in N_K} |\bar{\tau}_a / \tau_K - 1| \rightarrow 0$, then closes with the single sentence: “From the continuity assumed for the advection velocity $\mathbf{a}$ we have that $\bar{\tau}_a \rightarrow \tau_K$ as $h \rightarrow 0$.” But τ_1 depends on h through $c_1 \nu / h^2 + c_2 |\mathbf{a}| / h$, with the element length the continuous surrogate $\chi_1(\mathbf{x}_K, h)$, while the nodal $\bar{\tau}_a$ uses $\chi_1(\mathbf{x}_a, h)$. Writing $\rho = \chi_1(\mathbf{x}_a, h) / \chi_1(\mathbf{x}_K, h)$,

$$\frac{\bar{\tau}_{1,a}^{-1}}{\tau_{1,K}^{-1}} = \frac{c_1 \nu / (\rho h_K)^2 + c_2 |\mathbf{a}| / (\rho h_K)}{c_1 \nu / h_K^2 + c_2 |\mathbf{a}| / h_K} \xrightarrow{h \rightarrow 0} \begin{cases} \rho^{-2} & (\text{viscous-dominated}) \\ \rho^{-1} & (\text{convection-dominated}) \end{cases}$$

which equals 1 *only if* $\rho \rightarrow 1$. Continuity of $\mathbf{a}$ controls the amplitude $|\mathbf{a}(\mathbf{x}_a)| \rightarrow |\mathbf{a}|_{\infty, K}$ but says nothing about ρ . His Lemma 1 delivers only a *bounded* ratio $\chi_2/\chi_1 \leq \chi_0 < \infty$ (patch quasi-uniformity), and bounded is not $\rightarrow 1$. Indeed, even global quasi-uniformity with genuine size variation (element sizes in $[\underline{c}h, \bar{c}h]$, $\bar{c} > \underline{c}$) can keep ρ away from 1: a family with a fixed red/green refinement interface, refined uniformly, is shape-regular and patch-quasi-uniform with h -independent constants, yet the neighbour ratio stays ≈ 2 at the interface for every h , so in the viscous regime $|\bar{\tau}_a/\tau_K - 1| \approx |2^{-2} - 1| = 3/4 \not\rightarrow 0$. Thus Lemma 2 as written requires the element-size function to have *vanishing relative oscillation over each element*—a smooth-grading / asymptotic local quasi-uniformity property strictly stronger than nondegeneracy (H3), and not among H1–H6.

Note. The deduction is valid; only a hypothesis is left implicit, and the fix is one line (assume smooth grading, or restrict to globally quasi-uniform families, on which χ_1 is essentially constant and the term is trivial—for a uniform mesh χ_1 is constant and $\rho \equiv 1$). Codina’s convergence theorems, invoked on such meshes, are untouched. The porous smoothing lemma surfaces exactly this object: it decomposes $\Delta(\tau_1^{-1})$ into contributions from $\Delta\alpha$, $\Delta|\mathbf{a}|$, $\Delta(h^{-2})$, $\Delta(h^{-1})$ and introduces a modulus $\omega_\chi(h) \rightarrow 0$ for “the relative oscillation of $\chi_1(\cdot, h)$ over an element, which vanishes as $h \rightarrow 0$ under the smooth-grading assumption on the mesh family (bounded patch equivalence alone does not force it)”. In full honesty, the porous note flags $\omega_\chi(h) \rightarrow 0$ inside the proof rather than promoting it to its standing assumption list (A1); a maximally rigorous statement would make it a named hypothesis. This is a rigor-bookkeeping tightening, not a correctness bug: it should credit Codina’s construction, not frame it as broken.

(iii) The stability constants are conditional and disconnected from those used in practice (disclosure)

Codina’s Theorem 1 holds “for h sufficiently small and α in (27)–(28) large enough”. The proof exhibits the threshold only partially—it needs $\alpha > 3/2$ for one coercivity coefficient, plus positivity of several collected coefficients—and gives no consolidated admissible α_0 . His numerical section then fixes $c_1 = 4$, $c_2 = 2$ with no reconciliation to (27)–(28). These are self-consistent with $\alpha C_{\text{inv}} = 2$, i.e. $\alpha = 2/C_{\text{inv}}$; his own written step $\alpha > 3/2$ would then require $C_{\text{inv}} < 4/3$, which is not established and is generally false for higher-degree elements (where C_{inv} grows). So the stability/convergence theorems are correctly proved *conditional existence* results (“there exists a large enough amplification”), not validations of the $O(1)$ constants actually computed with.

Note. This is the standard, well-understood shape of stabilized/VMS a priori theory—provable stability needs stabilization above a threshold typically far larger than, and disconnected from, practice—and it is not a defect. The load-bearing point is one of *disclosure*: Codina 2007 does not flag the gap, whereas the porous OSGS analysis does, quantitatively. It exhibits an explicit admissible $\gamma_0 = 10(1 + M_2) (\geq 20)$ and computes that this demands $c_1 \gtrsim 400 C_{\text{inv}}^2$, $c_2 \gtrsim 20 C_{\text{inv}}$ —*well above* the experimental $c_1 = 4k^4$ (or $16k^4$), $c_2 = 2k^2$, “which are not shown to satisfy them.” It concludes that the result is “a conditional inf-sup stability theorem, not a calibration or validation of the stabilization constants employed in the computations.” It likewise records that the smallness threshold h_0 is not uniform in α_0 , ν , or the moduli of continuity of α and $\mathbf{a}$. None of this contradicts robust convergence at $c_1 = 4$: the sufficient conditions are explicitly non-sharp, so practice simply lies in the region the theory does not certify. Severity as a criticism of Codina: low; value of the explicit disclosure: high.

(iv) “Trivially $\beta_0 = 1$ ” and the weighted projection are exact only for globally constant τ (cleaner devices)

Two closely related points, both about *variable* τ , both handled more transparently by the porous analysis, neither an error.

(a) *The divergence slot.* Codina states that condition (35) “is trivially verified with $\beta_0 = 1$ ” for $z_h = \nabla \cdot \mathbf{v}_h$, because $\nabla \cdot \mathbf{v}_h$ has zero mean and therefore $\Pi_{\tau_2,0}(\nabla \cdot \mathbf{v}_h) = \Pi_{\tau_2}(\nabla \cdot \mathbf{v}_h)$. Testing the weighted projection against the constant shows only that the τ_2 -weighted integral is preserved, not the *unweighted* one; so for *variable* τ_2 with a continuous pressure space the constrained and unconstrained projections differ by one mean-mode and the sharp constant is $\beta_0 < 1$, not 1. (Exactness holds when τ_2 is *globally* constant, or when the pressure space is elementwise/discontinuous—not merely when τ_2 is elementwise-constant, which is already Codina’s case.) The consequence is benign: (35)/H4 still holds with a good $\beta_0 \in (0, 1]$, so his theorems are unaffected. The porous analysis reaches “constant exactly one” by a manifestly exact route instead: it projects $\nabla \cdot (\alpha \mathbf{u}_h)$ onto the *unconstrained* Q_h (an exact Pythagorean decomposition, no middle mode) and restores admissibility with a mean shift m_h of the pressure *test* function, shown invisible in every pairing (including the porous $\varepsilon(p_h, r_h)$ term). This is why the porous note imposes the projection-stability condition (A7) only on the *momentum* slot and needs no analogue for the divergence.

(b) *The weighted projection.* Codina’s a priori proof requires the τ -weighted projection (Remark 4: the weighted product is “essential” to the best-approximation property (46)), yet he presents the discrete matrix form only for globally constant τ_1 , where Π_{τ_1} collapses to the plain L^2 projection (M the ordinary Gram matrix). For variable τ the implemented (plain- L^2) and analyzed (weighted) methods differ, and the paper leaves this analyzed-versus-implemented distance implicit. The porous note makes it explicit (its “analyzed versus implemented” remark, item (i)): “the analysis uses the τ -weighted projections; the implementation uses the plain L^2 projection; the two coincide when τ_i is *globally* constant.”

Note. Both are cleaner or more explicit treatments, not corrections: Codina’s analyzed bilinear form openly uses Π_τ , his proof is correct for the method it analyzes, and his constant- τ matrix form is transparently flagged. The porous mean-shift device would equally sharpen the Oseen proof; $\varepsilon > 0$ and the α -weight only add cosmetic terms ($\int \nabla \cdot (\alpha \mathbf{u}_h) = 0$ still). Present these as variable-coefficient-robust reformulations that credit Codina’s Remark 1, not as audit findings.

A point that is *not* a gap: consistency framing

For completeness, one candidate concern turned out to be an overclaim and should be recorded as *agreement*. Codina’s phrase that dropping the viscous/force fluctuations “leads to a method which is still consistent (in a sense explained later; cf. Remark 3)” is sometimes read as loose. It is not. His Eq. (44) states outright that “the method is not consistent in the classical variational sense, since the RHS of (44) is not zero”; his Lemma 3 bounds the consistency error by $O(\varepsilon_C(h))$; and his Remark 3 exhibits the fully consistent *extended* form B_I^* with auxiliary unknowns to which “in a sense” refers. The porous note (`lem:consistency`) *transcribes* this—“not consistent in the classical variational sense (cf. Codina Eq. (44)); its consistency error is the value of the stabilization terms at the exact solution”—rather than sharpening it. (One small technical clarification: the Eq.-(44) non-consistency holds for *all* $k \geq 1$; the $k \geq 2$ threshold governs only the separately-neglected viscous-projection term, which is zero for P_1 because $\Delta \mathbf{u}_h = 0$.) The two analyses treat OSGS consistency identically and rigorously.

6 Proof-step ledger: where the two OSGS proofs diverge

Table 3 walks the two proofs in parallel and is the requested summary of “places in the OSGS proofs where there are differences worth mentioning.” Read it as: *same skeleton, four physical perturbations, one scope restriction, and a handful of expository tightenings*.

Table 3: Step-by-step ledger of the two OSGS proofs. F: forced by the new physics; T: an expository tightening of Codina 2007; S: a scope restriction.

Step	Codina 2007	Porous OSGS (difference)
Route	Non-coercive inf-sup with a special test function (Thm 1).	Same route; well-posedness obtained <i>from</i> the inf-sup estimate on the square system. F nothing—this is the inherited backbone.
Diagonal identity	$B_I(U_h, U_h) = \nu \ \nabla \mathbf{u}_h\ ^2 + \ \Pi_1^\perp \xi_h\ _{\tau_1}^2 + \ \Pi_2^\perp \delta_h\ _{\tau_2}^2$.	Adds $\ \sigma^{1/2} \mathbf{u}_h\ ^2$ and $\varepsilon \ p_h\ ^2$; viscous term becomes $2\nu \ \alpha^{1/2} \Pi^{\text{DS}} \nabla \mathbf{u}_h\ ^2$; requires $\nabla \cdot (\alpha \mathbf{a}) = 0$. F.
S_1, S_2	Project $\mathbf{a} \cdot \nabla \mathbf{u}_h + \nabla p_h$ and $\nabla \cdot \mathbf{u}_h$.	Project the <i>weighted</i> $\alpha X(U)$ and $\nabla \cdot (\alpha \mathbf{u})$; weighted projections Π_1, Π_2 . F.
Annihilation	Only viscous/force fluctuations neglected (zero for P_1).	Also: $\Pi_1^\perp(\sigma \mathbf{u}_h) = \mathbf{0}$, $\Pi_2^\perp(\varepsilon p_h) = 0$ for constant σ, ε (lem:annihilation)—the reaction/compressibility residuals are deleted <i>exactly</i> . F; the mechanism of the whole note.
τ_1, τ_2	Four free constants $c_1, \dots, c_4$; $\tau_2 = c_3 \nu + c_4 \mathbf{a} h$.	Reaction in the τ_1 denominator; τ_2 <i>slaved</i> to $\tau_{1,\text{NS}}$ (only c_1, c_2 free); identity (P3); dictionary $c_3 = 1, c_4 = c_2/c_1$ on reduction. F.
Smoothing	Lemma 2, $\psi(h) \rightarrow 0$ by continuity of $\mathbf{a}$.	Aggregates moduli of α , $\mathbf{a}$ and the mesh-grading ω_χ ; uses $\alpha_0 > 0$; surfaces the smoothing requirement (Section 5(ii)). F+T.
Patch equivalence	Lemma 1 / Eq. (31): equilibrates τ only.	Adds a <i>multiplicative</i> porosity equivalence (constant independent of the contrast α_∞/α_0); notes a multiplicative $ \mathbf{a} _\infty$ equivalence <i>fails</i> near zeros of $\mathbf{a}$. F.
Best-approximation	Eq. (46) for z ; Remark 4.	lem:bestapprox for the <i>product</i> αz ; Leibniz constant $C_{\alpha,r}, W^{m,\infty}$ porosity. F.
Interpolant	A single interpolant satisfying (30).	<i>Two</i> operators: boundary-preserving nodal (drives the H_0^1 global IBP in I2/I5) and Scott–Zhang of αz (drives the weighted best-approximation). F.
Triple norm / definiteness	$\nu^{1/2} \ \nabla \mathbf{v}\ + \dots$; definite by Poincaré.	Adds $2\nu \ \alpha^{1/2} \Pi^{\text{DS}} \nabla \mathbf{v}\ ^2$, $\ \sigma^{1/2} \mathbf{v}\ ^2$, $\varepsilon \ q\ ^2$; definiteness needs the Korn/ Π^{DS} identity and <i>full</i> Dirichlet data (lem:definiteness); dominates the ASGS norm term by term. F.
Special test function	$[\tau_1 \otimes \Pi_{\tau_1,0} \xi_h, \tau_2 \otimes \Pi_{\tau_2,0} \delta_h]$; both slots <i>constrained</i> ; $\beta_0 = 1$ “trivially” for the divergence.	Momentum slot constrained (needs (A7)); <i>pressure</i> slot uses the <i>unconstrained</i> projection + mean shift m_h , invisible in every term, making the mass-slot constant <i>exactly one</i> —no A7-analogue for the divergence. T (Section 5(iv-a)).

Table 3 (continued)

Step	Codina 2007	Porous OSGS (difference)
Perturbation terms	Viscous + two mixed terms; no reaction, no compressibility, no $\nabla\alpha$.	New pairings $T3 = \sigma(\mathbf{u}_h, \mathbf{w}_h)$ (absorbed by full D_σ), $T4 = \varepsilon(p_h, r_h)$ (absorbed by D_ε , needs $\varepsilon\tau_2 \leq C_2$); all inverse steps carry α ; T7 carries the $\nabla\alpha$ piece (factor M_2). F.
Collection / constants	“ $\alpha > 3/2$ ” suffices; no consolidated α_0 .	Explicit $\gamma_0 = 10(1 + M_2)$, $\psi(h_0) \leq \min\{1/8, \beta_0^2/16\}$; declared <i>conditional</i> , not a validation of $c_1 = 4k^4$ (Section 5(iii)). T.
Consistency	Eq. (44); mass part $\propto \Pi_2^\perp(\nabla \cdot \mathbf{u}) = 0$ <i>vanishes</i> .	Momentum part weighted, splits $\tau_1^{-1/2}$ (Da enters); <i>new</i> mass part $S_2(U, V_h) = -\varepsilon(\Pi_2^\perp p, \nabla \cdot (\alpha \mathbf{v}_h))_{\tau_2} \neq 0$. F (by ε).
Interpolation groups	Lemma 4: six groups.	Seven (I1–I7): new reactive I3 (via P6), compressibility I4 (via P5), plus the Damköhler factor entering the convective I2, compressibility I4 and pressure-gradient I5; I7 carries $\nabla\alpha$. F.
Error functional	Single ℓ^1 sum $\varepsilon_C(h)$, Eq. (36).	Sharp ℓ^2 $\Psi_O(h)$ with local $(c_1 + \text{Da}_{h,K})$; ℓ^1 majorant $\mathcal{E}(h)$ demoted, $h^{-d/2}$ loss flagged (Section 5(i)). T+F (Da).
Theorem + corollaries	Thm 2 (Method I); <i>also</i> Thms 3–4 (Method II) and 5–6 (viscous-dominated).	Only Method I proved. <i>New</i> corollaries: ASGS-norm domination up to $(c_1 + \text{Da}_{h,K})^{1/2}$, and the σ -robust optimal L^2 velocity estimate. Method II prospective only; viscous-dominated variant absent. F (corollaries) + S (Method II, Thms 5–6).
Reduction check	Thm 2 with $c_1 = \alpha^2 C_{\text{inv}}^2$, etc.	$\alpha \equiv 1, \sigma = \varepsilon = 0 \Rightarrow \text{Da}_{h,K} = 0$; $\Psi_O \rightarrow c_1^{1/2} \varepsilon_C$; recovers Codina Thm 2 under the dictionary $c_3 = 1, c_4 = c_2/c_1$. Consistency check.

7 Closing taxonomy

Every difference between the two OSGS analyses falls into exactly one of three classes. Table 4 is the one-page summary.

One-sentence conclusion. The porous OSGS a priori analysis is a faithful, strictly generalizing transcription of the Method I analysis of Codina 2007, whose every substantive deviation is forced by the porosity weight, the Darcy reaction, the artificial compressibility, or the deviatoric-symmetric viscous operator; where it appears to “correct” the ancestor it is in fact only sharpening exposition or disclosing a standard implicit convention, with the lone genuine—and entirely benign—finding being that the smoothing lemma of both papers tacitly assumes a smoothly graded mesh family.

References

- [1] G. Casas, J. González-Usúa, R. Codina, I. de-Pouplana, *A stabilized finite element method for incompressible, inertial flows in inhomogeneous porous media*, companion manuscript.

Class	Differences
F Forced by the new physics	Weighted residual αX ; weighted solenoidality $\nabla \cdot (\alpha \mathbf{a}) = 0$; porosity-resolution (A3) and patch-equivalence lemma; $\nabla \alpha$ term (factor M_2); deviatoric-symmetric energy and Korn definiteness; reaction diagonal and pairing T3; annihilation of the reactive/compressibility residual; the Damköhler identities (P5), (P6) and factor $(1 + \text{Da}_{h,K})^{1/2}$; τ_2 slaved to $\tau_{1,\text{NS}}$; compressibility pairing T4 and the <i>new non-vanishing</i> mass consistency term; interpolation groups I3, I4, I7; the isotropic-constant-coefficient restriction (A2); the σ -robust L^2 corollary.
T Tightening / disclosure of Codina 2007	The sharp ℓ^2 error functional with the $h^{-d/2}$ warning (Section 5(i)); surfacing the smooth-grading modulus ω_χ (Section 5(ii)); explicit conditional constants γ_0 and the theory-vs-practice disclosure (Section 5(iii)); the exact variable- τ mass-slot device and the explicit weighted-vs-plain projection distinction (Section 5(iv)); separate k_u, k_p and the stronger pressure regularity (A8).
S Scope restriction	Only Method I is transported. Method II is a prospective remark (proof not written); the pressure-only viscous-dominated variant (Codina Thms 5–6) has no porous analogue; the coefficients are frozen constant/isotropic and the analysis is linear (given $\mathbf{a}$, converged projection).

Table 4: Taxonomy of every catalogued difference. The overwhelming majority are F by the four physical ingredients of Equation (2); a compact set of T items sharpen Codina 2007’s exposition (only the smooth-grading one being a genuine, benign implicit hypothesis); the S items record what the porous analysis deliberately does not attempt.

- [2] *Stability and a priori error analysis of the OSGS variant* (self-contained companion note; identical to Appendix D of [1]), file `theory/osgs_a_priori/osgs_convergence.tex`.
- [3] R. Codina, *Analysis of a stabilized finite element approximation of the Oseen equations using orthogonal subscales*, Appl. Numer. Math. **58** (2008) 264–283 (online 18 Jan. 2007).
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- [5] R. Codina, J. Blasco, *A finite element formulation for the Stokes problem allowing equal velocity–pressure interpolation*, Comput. Methods Appl. Mech. Engrg. **143** (1997) 373–391.
- [6] L. R. Scott, S. Zhang, *Finite element interpolation of nonsmooth functions satisfying boundary conditions*, Math. Comp. **54** (1990) 483–493.